Multivariable Calculus @ JHU 2026

Lecture: 2

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Topics: Inner Product Spaces, Topology

Introduction:

In Lecture I we introduced the setting for our course, and discussed the natural types of mappings between such spaces. In this lecture, we will add some more structure to these spaces so that we can do geometry. We'll also use this machinery to understand the regularity of functions (eg: how continuous/differentiable they are), and the structure of sets of points in space (eg: how connected/complete they are). The basic ideas in this lecture will be used constantly in everything to come!

Section I: Inner Products

Let $(V, \mathbb{R})$ be an $\mathbb{R}$-vector space (although we could be more general and take $\mathbb{F}$-vector spaces instead...), which could be f.d. or infinite dimensional for the moment.

Definition 2.1: An *inner product* on V is a function

$$\langle -,- \rangle : V \times V \longrightarrow \mathbb{R}$$

satisfying the following rules $\forall\, u,v,w \in V$ and $\forall\, \mu, \lambda \in \mathbb{R}$:

1. **Symmetry** $\langle v, w \rangle = \langle w, v \rangle$
2. **Bilinearity** $\langle \mu v + \lambda w, u \rangle = \mu \langle v, u \rangle + \lambda \langle w, u \rangle$
3. **Positive Definiteness:** $\langle v, v \rangle \geq 0$ with "$=$" iff $v \equiv 0$.

Remark: To handle the case $\mathbb{F} = \mathbb{C}$, we need to impose some complex conjugations in the above rules; $\langle -, - \rangle : V \times V \to \mathbb{C}$ and

1. Conjugate Symmetry: $\langle v, w \rangle = \overline{\langle w, v \rangle}$
2. Linearity in 1st Argument: as above.
3. Positive Definiteness: as above.

Definition 2.2: An $\mathbb{F}$-vector space with a distinguished inner product is called an inner product space (IPS).

Key Example: In $\mathbb{R}^n$, we will frequently use the Euclidean inner product given as follows:

If $x = (x^1, \dots, x^n)$, $y = (y^1, \dots, y^n) \in \mathbb{R}^n$, then

$$\langle x, y \rangle_{euc} := x^1 y^1 + x^2 y^2 + \cdots + x^n y^n = \sum_{i=1}^n x^i y^i$$

We will use this inner product so much that eventually we will just write $\langle -, - \rangle_{euc} = \langle -, - \rangle$ when there is no ambiguity.

Exercise: Check that $\langle -, - \rangle_{euc}$ is indeed an inner product.

Remark: On $\mathbb{C}^n$, we would need to use $\langle x, y \rangle = \sum_{i=1}^n x^i \overline{y^i}$.

Definition 2.3: Let $(V, \mathbb{F})$ be a vector space. A norm on V is a function

$$\| - \| : V \to \mathbb{R}^{\geq 0}$$

satisfying the following rules for all $v, w \in V$, $\lambda \in \mathbb{F}$:

1. Positive 1-homogeneous: $\|\lambda v\| = |\lambda| \, \|v\|$
2. Positive Definite: $\|v\| \geq 0$ with "$=$" iff $v = 0$
3. Triangle Inequality: $\|v + w\| \leq \|v\| + \|w\|$.

A vector space with a distinguished norm is called a normed vector space. (NVS)

Key Example: In $\mathbb{R}^n$, we will frequently use the **Euclidean norm** given as follows: If $x = (x^1, \dots, x^n) \in \mathbb{R}^n$, then

$$\|v\|_{euc} = \left((x^1)^2 + (x^2)^2 + \cdots + (x^n)^2\right)^{\frac{1}{2}} = \left(\sum_{i=1}^n (x^i)^2\right)^{\frac{1}{2}}.$$

We will use this norm so much that eventually we will just write $\|-\|_{euc} = \|-\| \equiv |-|$ when there is no ambiguity.

Remark: On $\mathbb{C}^n$, we would need to use $\|v\| = \left(\sum_{i=1}^n x^i \overline{x^i}\right)^{\frac{1}{2}}$.

We'll check that this actually satisfies the axioms for a norm in a moment. To this end, notice that there is a connection between $\|-\|_{euc}$ and $\langle -, - \rangle_{euc}$:

$$\|v\|_{euc} = \sqrt{\langle v, v \rangle_{euc}} \qquad \text{for all } v \in \mathbb{R}^n.$$

In fact:

Proposition 2.4: Every inner product induces a norm via the formula

$$\|-\| = \langle -, - \rangle^{\frac{1}{2}}.$$

Hence, every IPS is a NVS as well.

The following result is indispensible, and is useful in proving the above proposition:

Proposition 2.5: (Cauchy-Schwarz Inequality) (Let $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$ here)

If $(V, \langle -, - \rangle)$ is an IPS, and $\|-\|$ is the induced norm, then

$$|\langle v, w \rangle| \leq \|v\| \, \|w\| \qquad \text{for every } v, w \in V,$$

with "=" iff v and w are scalar multiples of one another.

Proof: If $w = 0$, the result is trivial, so assume $w \neq 0$. Then if $\lambda \in \mathbb{F}$ is arbitrary,

$$0 \leq \|v - \lambda w\|^2 = \langle v - \lambda w, v - \lambda w \rangle = \langle v, v \rangle - \lambda \langle w, v \rangle - \overline{\lambda} \langle v, w \rangle + \lambda \overline{\lambda} \langle w, w \rangle$$
$$= \|v\|^2 - \left(\overline{\lambda \langle w, v \rangle} + \lambda \langle w, v \rangle\right) + |\lambda|^2 \|w\|^2$$
$$= \|v\|^2 - 2 \operatorname{Re}(\lambda \langle w, v \rangle) + |\lambda|^2 \|w\|^2$$

Now, since $\lambda \in \mathbb{F}$ was arbitrary, we can see what happens if we choose

$$\lambda = \frac{\overline{\langle w, v \rangle}}{\|w\|^2} : \quad \text{(here is where we use } w \neq 0\text{)}$$

$$0 \leq \|v\|^2 - 2\,\mathrm{Re}(\lambda \langle w, v \rangle) + |\lambda|^2 \|w\|^2 = \|v\|^2 - 2\frac{|\langle w, v \rangle|^2}{\|w\|^2} + \frac{|\langle w, v \rangle|^2}{\|w\|^4}\|w\|^2$$

$$= \|v\|^2 - \frac{|\langle v, w \rangle|^2}{\|w\|^2}.$$

Rearranging this gives the C-S inequality, and clearly "=" in C-S implies that $\langle v - \lambda w, v - \lambda w \rangle = 0 \iff v = \lambda w$ for λ as above, by the inner product axioms.



Proof of Proposition 2.4: We need to check that the "induced norm" $\|-\| = \sqrt{\langle -, - \rangle}$ satisfies the norm axioms for real: Let $v, w \in V$, $\lambda \in \mathbb{F}$:

1. $\|\lambda v\| = \sqrt{\langle \lambda v, \lambda v \rangle} = \sqrt{\lambda \bar{\lambda} \langle v, v \rangle} = |\lambda| \sqrt{\langle v, v \rangle} = |\lambda| \|v\|$
2. $\|v\| = \sqrt{\langle v, v \rangle} \geq 0$ with "=" iff $v = 0$.
3. Δ-ineq: $\|v + w\|^2 = \langle v + w, v + w \rangle = \|v\|^2 + \langle v, w \rangle + \langle w, v \rangle + \|w\|^2$

$$= \|v\|^2 + 2\,\mathrm{Re}\langle v, w \rangle + \|w\|^2$$

$$\leq \|v\|^2 + 2|\langle v, w \rangle| + \|w\|^2$$

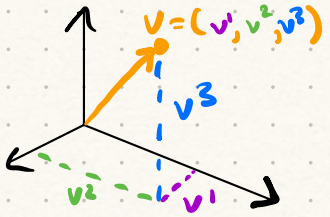
$$(\text{C-S}) \leq \|v\|^2 + 2\|v\|\|w\| + \|w\|^2 = (\|v\| + \|w\|)^2$$



Intuition: It's easiest to understand the choice of λ in the proof of C-S by considering the $\mathbb{R}$-case: First we seek to understand how close v and w can be up to scaling: this tells us to consider the quantity $0 \leq \|v - \lambda w\|^2 = \|w\|^2 \lambda^2 + 2\langle v, w \rangle \lambda + \|v\|^2$. If we view the RHS as a polynomial in λ (for fixed v, w), then this non-negative quadratic function must attain a non-negative minimum value somewhere, hence we seek all possible $\lambda \in \mathbb{R}$ so that $p(\lambda) = \|w\|^2 \lambda^2 + 2\langle w, v \rangle \lambda + \|v\|^2$ is minimized: setting $p'(\lambda) = 0$ we immediately find the value $\lambda = \frac{\langle v, w \rangle}{\|w\|^2}$.

A key Geometric Picture: In any vector space, a norm $\|-\|$ and inner product bestow the world with geometry:

- The norm gives us a way to measure the size/length of a vector. Indeed if $v=(v^1,\dots,v^n)\in\mathbb{R}^n$, the expression $\|v\|_{euc}=\|v\|_2$ is nothing other than the Pythagorean Theorem's expression for the length of v:



$$\|v\|^2=(v^1)^2+(v^2)^2+(v^3)^2$$

Other choices of norms simply capture alternative ideas of size/distance.

- The inner product gives us a way to measure how similar two vectors are. Recalling Cauchy-Schwarz, let v,w both be nonzero, and recall that then
$$-1\le\frac{\langle v,w\rangle}{\|v\|\,\|w\|}\le 1.$$
Consequently, there is a unique angle $\theta\in[0,\pi]$ so that
$$\frac{\langle v,w\rangle}{\|v\|\,\|w\|}=\cos\theta.$$

This angle θ has an important geometric meaning: writing
$$\left\langle\frac{v}{\|v\|},\frac{w}{\|w\|}\right\rangle=\cos\theta$$
and drawing $v/\|v\|$ and $w/\|w\|$ as unit-length vectors gives way to the following sort of pictures (notice how the sign of $\cos\theta$ tells us if $\frac{v}{\|v\|}$ and $\frac{w}{\|w\|}$ point "with" or "against" each other):



In particular, if $\theta=\pi/2$ then v and w are at right angles to each other and we say that v,w are orthogonal. If $\theta=0$ or π, then we are in the equality case of Cauchy-Schwarz and the vectors are dependent.

Note: To actually prove this geometric picture in $\mathbb{R}^n$ with $\langle-,-\rangle_{euc}$ and $\|-\|_{euc}$, one can first apply rotation matrices to $\mathbb{R}^n$ (which preserve angles, norms) to put $\frac{v}{\|v\|}$ and

$\frac{w}{\|w\|}$ into $\mathbb{R}^2 \times \{\vec{0}^{n-2}\}$. Thus, WLOG, we only need to verify the geometric picture for $\frac{v}{\|v\|}, \frac{w}{\|w\|} \in \mathbb{R}^2$. Writing $v = (\cos\theta_v, \sin\theta_v)$ and $w = (\cos\theta_w, \sin\theta_w)$ for $\theta_v, \theta_w \in [0, 2\pi)$, we have

$$\left\langle \frac{v}{\|v\|}, \frac{w}{\|w\|} \right\rangle = \cos\theta_v \cos\theta_w + \sin\theta_v \sin\theta_w = \cos(\theta_v - \theta_w)$$

and clearly $\theta_v - \theta_w$ is the angle between v and w. There is a unique $\theta \in [0, \pi]$ so that $\cos(\theta) = \cos(\theta_v - \theta_w)$ (use evenness of $\cos(-)$ and 2π-periodicity), and we take this θ to denote the angle.

We end this section with a few remarks, and some important facts about norms:

As you might suspect, there are in fact **many** different norms which exist on a vector space — not just the one that comes from the inner product. Even still, there are vector spaces that have norms, but do not have inner products.(*) In any case, occasionally other choices of norms can be useful: for instance we will sometimes find it convenient to use:

- the **sup-norm**, or **ℓ^∞-norm**, on $\mathbb{R}^n$: $\|v\|_\infty = \|(v^1, \dots, v^n)\|_\infty = \max_{1 \le i \le n} |v^i|$ ($|\cdot|$ = abs. value in $\mathbb{F}$)
- the **ℓ^p-norm**, where $p \in [1, \infty)$, on $\mathbb{R}^n$: $\|v\|_p = \left(\sum_{i=1}^n |v^i|^p\right)^{\frac{1}{p}}$ (note $\|\cdot\|_{\text{euc}} \equiv \|\cdot\|_2$)
- If $M \in \mathcal{M}^{n \times m}(\mathbb{F})$ is an $(n \times m)$-matrix with entries in $\mathbb{F}$, then any of these norms can be applied to M: just associate to M the $(nm \times 1)$-column vector of all its entries and apply the norm to this vector.

Fortunately, for many purposes it doesn't matter which norm we use:

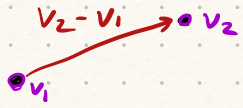
Theorem 2.6: Let $(V, \mathbb{F})$ be any **finite dimensional** vector space, and $\|\cdot\|_1$ and $\|\cdot\|_2$ any two norms on V. Then these norms are **equivalent**: there are constants $C, D \in (0, \infty)$ fixed so that $\|x\|_1 \le C\|x\|_2$, $\|x\|_2 \le D\|x\|_1$ for all $x \in V$. This can fail in **infinite** dimensional vector spaces!

(*) There is a famous theorem of Jordan-Von-Neumann, that says that if a norm $\|\cdot\|$ on a vector space satisfies the **parallelogram law** $\|x+y\|^2 + \|x-y\|^2 = 2(\|x\|^2 + \|y\|^2)$, then $\|\cdot\|$ must come from some inner product $\langle -, - \rangle$ on the vector space.

Section II: Metric Space Topology

Because $(\mathbb{R}^n, \|-\|)$, as a vector space with a norm, has a notion of size, we can talk about the distances between pairs of vectors:

Distance from v_1 to $v_2 = \|v_1 - v_2\|$.



This is a basic example of a metric space:

Definition 2.7: Given a set X, a metric on X is a function $d: X \times X \to \mathbb{R}^{\geq 0}$ such that the following rules hold for all $x, y, z \in X$:

1) Symmetry: $d(x,y) = d(y,x)$
2) Positive Definite: $d(x,y) \geq 0$, with "=" iff $x = y$
3) Triangle Inequality: $d(x,z) \leq d(x,y) + d(y,z)$

The pair (X, d) is called a metric space. If $Y \subseteq X$, then the restriction of d to $Y \times Y \subseteq X \times X$ makes (Y, d) into a metric space of its own, and we call it a metric subspace with the restriction metric.

Key Example (already noted): The set $\mathbb{R}^n = \{(x^1, \dots, x^n) : x^i \in \mathbb{R}\}$ becomes a metric space when endowed with the Euclidean distance $d_{euc}(x,y) = \sqrt{\sum_{i=1}^n |x^i - y^i|^2}$. Of course, we see that $d_{euc}(x,y) = \|x - y\|_{euc}$, and this makes it easy to check the metric space axioms.

Key Point (for us): Thinking of $\mathbb{R}^n$ also as a metric space allows us to discuss convergence, and ideas related to it. In particular, it lets us talk about concepts like continuity for functions, which is what we turn to next. We opt for giving the minimal amount of theory necessary for what we need.

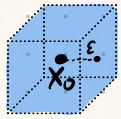
As a general remark: The mathematical field of Topology concerns the topic of continuity, and the invariants associated to it. At its core, the ideas of this field are built out of the construction of a topology on a space of points: a distinguished collection of subsets of the space which

group points together, letting us which groups of points are considered to be "related to each other". In general, these distinguished subsets of the space in the topology are called **open sets**. In the setting of $(\mathbb{R}^n, d_{euc})$, these distinguished open sets arise in the following way:

Definition 2.8: Given $x_0 \in \mathbb{R}^n$ and $\varepsilon > 0$, we call



- $B_\varepsilon(x_0) = \{x \in \mathbb{R}^n : d(x, x_0) < \varepsilon\}$ an **open ball of radius $\varepsilon > 0$ about x_0.**



- $Q_\varepsilon(x_0) = (x_0^1 - \varepsilon, x_0^1 + \varepsilon) \times \cdots \times (x_0^n - \varepsilon, x_0^n + \varepsilon)$ an **open cube of radius $\varepsilon > 0$ about x_0.**

Either of these types of sets can form the **building blocks** for the **standard topology** of **open sets** in $(\mathbb{R}^n, d_{euc})$, in the following sense:

Definition 2.9: A subset $U \subseteq \mathbb{R}^n$ is **open** if for every $x \in U$, we can find **some** $\varepsilon > 0$ so that $B_\varepsilon(x) \subseteq U$ (or alternatively that $Q_\varepsilon(x) \subseteq U$).



A subset $C \subseteq \mathbb{R}^n$ is said to be **closed** if $\mathbb{R}^n \setminus C$ is open.

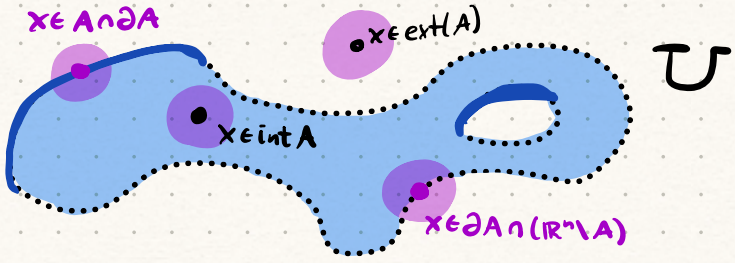
Exercise: Show that, indeed, $B_\varepsilon(x)$ and $Q_\varepsilon(x)$ are themselves open sets for every $\varepsilon > 0$. Also show that the set $[a^1, b^1] \times \cdots \times [a^n, b^n]$ is closed in $(\mathbb{R}^n, d_{euc})$ (this justifies calling this sort of set a **closed rectangle**)

Definition 2.10: Let $A \subseteq \mathbb{R}^n$ be any subset, and $x \in \mathbb{R}^n$ any point. Then exactly one of the following occurs:

1) There is an open ball (cube) $B_\varepsilon(x) \subseteq A$. We say a **is in the interior of** A, and write $a \in \text{int}(A)$. The set $\text{int}(A)$ is open.

2) There is an open ball $B_\varepsilon(x) \subseteq \mathbb{R}^n \setminus A$. We say a is in the exterior of A, and write $a \in \text{ext}(A)$. The set $\text{ext}(A)$ is open.

3) For every $\varepsilon > 0$, $B_\varepsilon(x)$ intersects both A and $\mathbb{R}^n \setminus A$. We say a is in the boundary of A, and write $x \in \partial A$. The set ∂A is closed.



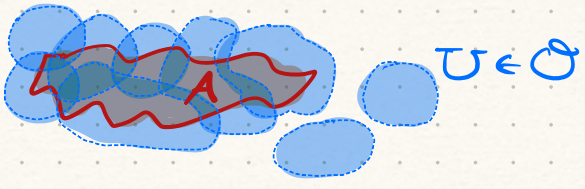
Examples: ① $A = B_\varepsilon(x_0)$: $\text{int}(A) = A$, $\text{ext}(A) = \{x \in \mathbb{R}^n : d(x, x_0) > \varepsilon\}$, $\partial A = \{x \in \mathbb{R}^n : d(x, x_0) = \varepsilon\}$ = sphere of radius ε at x_0.

② $A = \partial B_\varepsilon(x_0)$: $\text{int}(A) = \emptyset$, $\text{ext}(A) = \{x \in \mathbb{R}^n : d(x, x_0) > \varepsilon \text{ or } < \varepsilon\}$, $\partial A = A$.

③ $A = \{x = (x^1, \dots, x^n) \in \mathbb{R}^n : x^i \in \mathbb{Q}\}$: $\text{int}(A) = \emptyset$, $\text{ext}(A) = \emptyset$, $\partial A = \mathbb{R}^n$.

Aside from open and closed sets, there is another class of sets that will be very important for us in the future: compact sets. First we introduce

Definition 2.11: Let $A \subseteq \mathbb{R}^n$ be any subset, and let $\mathcal{O}$ be a collection of open sets. Then $\mathcal{O}$ is an open cover of A iff $A \subseteq \bigcup_{U \in \mathcal{O}} U$:



A is said to be compact if every open cover $\mathcal{O}$ of A has a finite subfamily which is still an open cover. (ie non-compact sets have at least one open cover which truly must contain an infinite number of open sets in order to cover all of A).

Examples: ① $\mathbb{R}$ is non-compact, as $\mathbb{R} \subseteq \bigcup_{n=1}^{\infty} (-\infty, n)$.

② Any finite set $\{p_1, \dots, p_k\} \subseteq \mathbb{R}^n$ is compact (any open cover needs at most k open sets).

③ The infinite set $A = \{0\} \cup \{\frac{1}{n}\}_{n=1}^{\infty} \subseteq \mathbb{R}$ is compact: if $\mathcal{O}$ is an open cover of A and $U \in \mathcal{O}$ contains $0 \in A$, then $\exists \varepsilon > 0$ so that $B_\varepsilon(0) \subseteq U$ by definition of being open. Thus, for every $n \geq \frac{1}{\varepsilon}$, $\frac{1}{n} \in B_\varepsilon(0) \subseteq U$. We thus only need at most $\lfloor \frac{1}{\varepsilon} \rfloor$ other open sets in $\mathcal{O}$ to cover the points $1, \frac{1}{2}, \dots$, not in U.

④ Any open set $A \subseteq \mathbb{R}^n$ is non-compact: For a truly infinite open cover, take $\mathcal{O} = \{ B_{\frac{f(x)}{2}}(x) : x \in A \}$, where $f(x) = \text{dist}(x, \partial A)$.

Fortunately, there is a beautiful criterion for compactness in topological spaces like $\mathbb{R}^n$:

Theorem 2.12: (Heine-Borel Theorem)
A set $A \subseteq \mathbb{R}^n$ is compact iff it is ① closed and ② bounded.

Of course, a bounded set is any set which can be covered by a single ball of finite radius: $A \subseteq B_R(x_0)$ for some x_0 and some $R < \infty$.

We won't prove the Heine-Borel Theorem in class, but it follows the following steps:

① Show all closed, bounded intervals $[a,b] \subseteq \mathbb{R}$ are compact.

Hint: Given any open cover $\mathcal{O}$ of $[a,b]$, let $\alpha = \sup\{x \in [a,b] : [a,x] \text{ is covered by a finite collection of sets in } \mathcal{O}\}$. Show $\alpha = b$.

② Show products of compact sets are compact: $A \subseteq \mathbb{R}^n$, $B \subseteq \mathbb{R}^m$ compact $\Rightarrow A \times B \subseteq \mathbb{R}^{n+m}$ is compact.

Hint: Show that if $B \subseteq \mathbb{R}^m$ is compact, $x \in \mathbb{R}^n$, and $\mathcal{O}$ is an open cover of $\{x\} \times B$, then there is a single open set $U \subseteq \mathbb{R}^n$ containing x so that $U \times B$ has a finite cover contained in $\mathcal{O}$.



③ Prove the Theorem

Hint: If $A \subseteq \mathbb{R}^n$ is closed/bounded, find a closed box $B \supseteq A$, which is compact by ②. Then, if $\mathcal{O}$ is an open cover of A, note that $\mathcal{O} \cup \{\mathbb{R}^n \setminus B\}$ is an open cover of B...

Section III: Functions from $\mathbb{R}^n$ to $\mathbb{R}^m$ and Continuity

We are now ready to discuss the main objects of analysis in our course:

Definition 2.13: A function $f:\Omega\subseteq\mathbb{R}^n\to\mathbb{R}^m$ is a rule which associates to each $x\in\Omega$ (the *domain* of f) a unique point called $f(x)\in\mathbb{R}^m$. We call $\mathbb{R}^m$ the *codomain* of f, and the subset $f(\Omega):=\{f(x)\in\mathbb{R}^m : x\in\Omega\}$ the *image* of f.

One of the most common ways of writing a "vector valued function" as above is in terms of its component functions: If $f:\Omega\subseteq\mathbb{R}^n\to\mathbb{R}^m$ then for every $x\in\Omega$ we can write

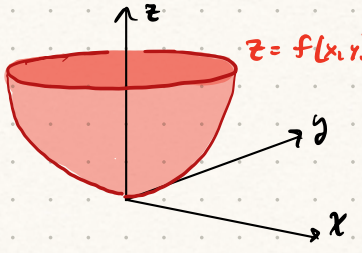
$$f(x)=f(x^1,\dots,x^n)=\left(f^1(x^1,\dots,x^n),\dots,f^m(x^1,\dots,x^n)\right)=\left(f^1(x),\dots,f^m(x)\right)$$

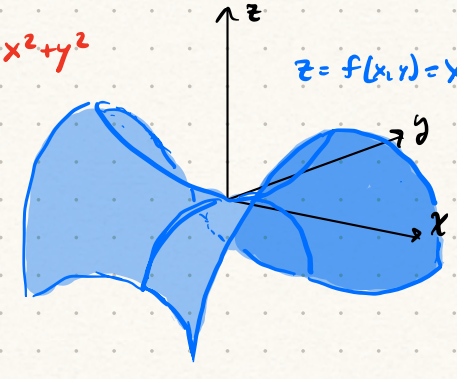
hence, $f:\Omega\to\mathbb{R}^m$ determines m component functions $f^i:\Omega\to\mathbb{R}$ and vice-versa. The "algebra" of vector valued functions follows the same sort of rules as usual functions, using the vector space structure of $\mathbb{R}^m$:

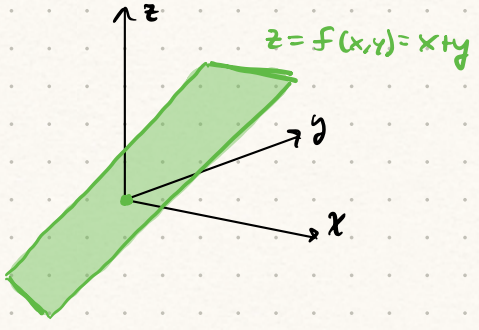
1. **Addition:** If $f,g:\Omega\to\mathbb{R}^m$, then $(f+g)(x):=f(x)+g(x)=\left(f^1(x)+g^1(x),\dots,f^m(x)+g^m(x)\right)$
2. **Multiplication:** If $f:\Omega\to\mathbb{R}^m$, $g:\Omega\to\mathbb{R}$, $(g\cdot f)(x)=g(x)f(x)=\left(g(x)f^1(x),\dots,g(x)f^m(x)\right)$.
3. **Composition:** If $f:\Omega\to\mathbb{R}^m$, $g:\tilde{\Omega}\subseteq\mathbb{R}^m\to\mathbb{R}^k$ with $f(\Omega)\subseteq\tilde{\Omega}$, $(g\circ f)(x)=g(f(x))$.
4. **Preimages:** If $y\in\mathbb{R}^m$, $f^{-1}(y)=\{x\in\Omega : f(x)=y\}$ is called the *preimage* of y.

   If f is injective ($f(x)=f(y)$ iff $x=y$), we have a well defined *inverse function* $f^{-1}:f(\Omega)\to\Omega$.

It's often very helpful to think of a function geometrically in terms of its graph, especially when $f:\mathbb{R}^2\to\mathbb{R}$, since we can draw pictures like:







Just as in the single variable case, we define continuity of a vector valued function so as to capture the idea that making $x \in \Omega$ close enough to $a \in \Omega$ ensures that $f(x)$ is close to $f(a)$. Rigorously, we define (using now the metric space structure of $\mathbb{R}^n$ coming from the Euclidean norm $\|-\| \equiv \|-\|_{euc}$):

Definition 2.14: $f: \Omega \to \mathbb{R}^m$ is *continuous at a point* $a \in \Omega$ iff $\lim_{\Omega \ni x \to a} f(x) = f(a)$, meaning that $\forall \delta > 0$, $\exists \varepsilon = \varepsilon(\delta) > 0$ such that $\|x - a\| < \varepsilon$, $x \in \Omega$, implies $\|f(x) - f(a)\| \leq \delta$. f is *continuous on* Ω iff it is continuous at every $a \in \Omega$.

Sometimes, it is more useful to use the following equivalent definition (which actually is the *general* definition of continuity for maps in general topological spaces):

Definition 2.15: If $f: \Omega \subseteq \mathbb{R}^n \to \mathbb{R}^m$, then we say f is continuous on Ω iff for every open set $U \subseteq \mathbb{R}^m$, $f^{-1}(U) = V \cap \Omega$ for some open set $V \subseteq \mathbb{R}^n$. Note that, in particular, if $\Omega \subseteq \mathbb{R}^n$ is open, this is equivalent to simply asking that $f^{-1}(U)$ be open in $\mathbb{R}^n$.

Here are some of the *key facts* about continuous functions $f: \Omega \subseteq \mathbb{R}^n \to \mathbb{R}^m$:

Proposition 2.16:

1. All linear maps $T: \mathbb{R}^n \to \mathbb{R}^m$ are continuous.

   Hint: Show that for all $x \in \mathbb{R}^n$, $\|T(x)\| \leq \|T\| \cdot \|x\|$.

2. Any composition, addition, or multiplication of continuous functions is also continuous.

   Hint: Show $s(x,y) = x + y$, and $p(x,y) = xy$, are continuous. Then compose.

3. $f = (f^1, \ldots, f^m)$ is continuous iff each f^i is.

   Hint: Let $\pi^i: \mathbb{R}^n \to \mathbb{R}$, $x = (x^1, \ldots, x^n) \mapsto x^i$ be the ith coordinate projection. Show that π^i is linear, and note that $f^i = \pi^i \circ f$.

4. If $f: \Omega \to \mathbb{R}^m$ is continuous and $\Omega \subseteq \mathbb{R}^n$ is compact, then so is $f(\Omega) \subseteq \mathbb{R}^m$.

   Hint: Pull back an open cover of $f(\Omega)$, and use Definition 3.15

5. If $f: \Omega \to \mathbb{R}$ is continuous and $\Omega \subseteq \mathbb{R}^n$ is compact, then f attains a maximum and minimum value on Ω.

   Hint: $f(\Omega) \subseteq \mathbb{R}$ is compact, so one may apply Heine-Borel.

Exercises for Lecture II

1. Check $\|\cdot\|_{\text{Euc}}$ and $\langle \cdot, \cdot \rangle_{\text{Euc}}$ satisfy their axioms
2. Prove reverse Δ-ineq
3. Prove Pythagorean Theorem
4. Show linear $T: \mathbb{R}^n \to \mathbb{R}^m$ is cts
5. Dual vector space?

Metric Spaces:

1. Prove balls, cubes open
2. Int, Ext, ∂ of sets

To Do:

- CS inequalities
- $\square$ law
- $\perp$
- equiv norms, other inner products
- metric spaces
- topology, open, closed, compact, connected
- limits, continuity